Birth Order and Academic Productivity in Pre-Modern Europe: Evidence from Brothers in Universities and Learned Academies

Silvia Busquets Xapellí

Dissertation written under the supervision of Professor Hugo Reis and Professor David De la Croix

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Abstract

This paper examines the influence of birth order on scholarly productivity among historical European scholars. Using a new dataset of 645 brother pairs from the Repertorium Eruditorum Totius Europae (RETE), we compare the academic output of older and younger siblings within families. The empirical strategy relies on within-family comparisons, helping to control for shared family characteristics. Scholarly productivity is measured using bibliographic indicators from the Virtual International Authority File (VIAF) together with supplementary measures of academic visibility. The results indicate a modest productivity advantage for older brothers: they are more likely to have at least one recorded publication and tend to display slightly higher levels of scholarly output. However, conditional on publishing, productivity differences become substantially smaller. The results also reveal substantial historical heterogeneity. Before the sixteenth century, productivity differences are more strongly associated with sibling age gaps, whereas after the sixteenth century differences in institutional affiliation become more strongly associated with productivity divergence between brothers. Overall, the findings suggest that differences in institutional affiliation are more consistently associated with within-family productivity divergence than regional inheritance systems.

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Resumo em português

Este artigo analisa a influência da ordem de nascimento na produtividade académica, em termos históricos, entre académicos europeus. Utilizando um novo conjunto de dados com 645 pares de irmãos do Repertorium Eruditorum Totius Europae (RETE), comparamos a produção académica de irmãos mais velhos e mais novos no seio da mesma família. A estratégia empírica assenta em comparações intrafamiliares, permitindo controlar características familiares comuns aos irmãos. A produtividade académica é medida através de indicadores bibliográficos do Virtual International Authority File (VIAF), complementados por medidas adicionais de visibilidade académica. Os resultados apontam para uma vantagem moderada dos irmãos mais velhos: apresentam maior probabilidade de ter pelo menos uma publicação registada e tendem a revelar níveis ligeiramente superiores de produção académica. No entanto, condicionando a análise à existência de publicação, as diferenças de produtividade tornam-se substancialmente mais reduzidas. Os resultados revelam ainda uma heterogeneidade histórica significativa. Antes do século XVI, as diferenças de produtividade estão mais associadas às diferenças etárias entre irmãos; após o século XVI, as diferenças de afiliação institucional passam a estar mais fortemente relacionadas com a divergência de produtividade académica. De forma geral, os resultados sugerem que os contextos institucionais desempenharam um papel mais relevante do que os sistemas de herança na explicação das diferenças de produtividade intrafamiliares entre académicos.

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Declaration on the Use of Artificial Intelligence

During the preparation of this thesis, I made use of generative artificial intelligence tools as a technical aid. Specifically, these tools were used to assist with writing and debugging code in R (RStudio) and to help correct grammatical and syntactic errors in the text.

The use of these tools was strictly limited to a supporting role. The research question, theoretical framework, empirical strategy, construction of the dataset, analysis, interpretation of the results, and conclusions are entirely my own work. All outputs produced with the assistance of these tools were carefully reviewed, verified, and edited by me, and I take full responsibility for the content of this thesis.

1. Introduction

There is a large body of literature that examines the effects of birth order. For economists in particular, research shows that whether you are the oldest or youngest or in between is not just a family curiosity but a factor with a distinct effect on education, skill development, and life success.

Empirical evidence using late twentieth- and early twenty-first century administrative data from Norway, the United States, and the United Kingdom consistently documents a firstborn premium, i.e., that firstborns are systematically better off than their later-born siblings with respect to school and other life outcomes (Devereux et al., 2005; Behrman and Taubman, 1986; Booth and Kee, 2009). The reasons for this phenomenon are mainly explained by the allocation of time and money, as well as different expectations set for each child. That is why it is important to understand birth-order effects: they are the way we see families transferring advantage or disadvantage from one generation to the next.

However, evidence from historical populations in Europe suggests that birth-order effects may be limited or context-dependent. Using a dataset covering England between 1680 and 2024, Clark and Cummins (2024) find a small impact of birth order on social outcomes in the population over the long run.

Using Canadian survey data and U.S. administrative and educational data from the late twentieth century, recent studies comparing siblings within the same family show that younger children receive lower educational investments and attain lower levels of education than their older siblings (Kantarevic and Mechoulan, 2006; Hotz and Pantano, 2015).

Several mechanisms have been proposed to explain these patterns. One explanation is the dilution of parental resources in larger families. Another explanation is different parental investment and attention across siblings according to birth order. An additional explanation is sibling competition (Becker and Tomes, 1976; Blake, 1981; Price, 2008).

However, there is a gap in our historical understanding, since it is unclear whether these effects were also prevalent during pre-industrial times in Europe (from the thirteenth to the eighteenth centuries), as society was different from the contemporary one. In particular, families were larger, child mortality was higher, and legal norms, particularly those on inheritance, varied across regions. For these reasons, the impact of birth order in these historical contexts, and especially among elite groups such as scholars and intellectuals, remains an open question.

This dissertation focuses on this gap by looking at birth order effects amongst European scholars active in Europe during the aforementioned period. We combine biographical data on European scholars from the Repertorium Eruditorum Totius Europae (RETE), covering both universities and learned academies across Europe, with information on family relationships and regional inheritance systems. Focusing on scholars makes it possible to measure long-run academic success through publications and scholarly visibility instead of short-term educational outcomes.

One of the key components of this work is the role of institutional environments and inheritance systems. Inheritance institutions varied substantially across European regions between the sixteenth and eighteenth centuries (Todd, 1990). In Europe, there were primogeniture systems and egalitarian systems. Primogeniture systems concentrate family assets primarily on the eldest son, as opposed to egalitarian inheritance regimes that divide assets equally among siblings. These differences likely affect parental investment decisions and sibling competition. Recent research using historical data from France and other European regions between the eighteenth and twentieth centuries suggests a link between family systems (including inheritance institutions) and long-run demographic/socioeconomic outcomes (Duranton et al., 2009; Gobbi et al., 2026). However, it remains unclear to what extent these institutions generated inequalities within scholar families themselves.

Using new data on pairs of brothers who were both scholars, we pursue three main objectives. First, to determine if older brother scholars consistently outperformed their younger brothers and, if so, whether the size of this difference is similar to that found in modern studies. Second, to examine whether observed differences can be explained by factors such as age gaps and family background, and in particular whether birth-order productivity differences persist within shared institutional environments or instead become larger when siblings go to different institutions. Third, to examine how institutional environments and inheritance systems are associated with birth-order productivity differences within families. In particular, we hypothesize that the firstborn advantage will have been more pronounced in non-egalitarian systems.

The empirical strategy consists of within-family comparisons between brothers for whom birth order can be established from recorded birth years. This approach helps control for many characteristics shared within families, including parental background, social environment, and other family-level characteristics. Brother pairs with identical recorded birth years are excluded from the birth-order analysis, since older and younger siblings cannot be clearly identified in such cases. We measure scholarly productivity using publications and other historical indicators of academic visibility. We then examine how these outcomes relate to birth order and regional inheritance systems. The empirical approach combines paired comparisons with regression analysis.

This dissertation presents new evidence on the relationship between birth order and scholarly productivity among historical European scholars. We utilize a new dataset of 645 brother pairs to compare the performance of older and younger siblings, controlling for age gap, family background, and institutional context.

The findings provide evidence of a modest older-brother advantage in observed scholarly productivity among historical European scholars. However, this effect operates primarily through the extensive margin of academic activity. Older brothers are substantially more likely to have at least one recorded publication, whereas, conditional on publishing, productivity differences between siblings become small and statistically insignificant.

Using measures derived from VIAF, paired comparisons show that older brothers produced, on average, approximately two additional titles and tended to perform better across several indicators of academic prominence.

Last, we investigate whether inheritance systems and institutional environments are associated with different patterns of birth-order productivity gaps. The results suggest that institutional environments play a more robust role than inheritance systems in explaining productivity divergence between scholarly siblings. In particular, productivity differences become larger when brothers are affiliated with different academic institutions.

This dissertation contributes to several strands of the literature. First, it extends the literature on birth-order effects into the context of pre-industrial European elites by providing new evidence on scholarly productivity between the thirteenth and eighteenth centuries. Second, by combining RETE biographical data with information on family relationships, institutional environments, and inheritance systems, it contributes to the study of how historical academic environments shaped intra-family inequality in Europe. Third, the analysis highlights that birth-order differences among scholars operate mainly through entry into scholarly production rather than conditional productivity, thereby distinguishing between the extensive and intensive margins of intellectual activity.

The paper is organized as follows. Section 2 reviews the relevant literature on birth order, human capital formation, and family institutions. Section 3 describes the data, outlines the empirical strategy, and presents the main results together with their interpretation and limitations. Section 4 concludes.

2. Literature Review

In this section, we overview the key literature on birth order effects among historical scholars. We will look at four main categories. We will first summarise what is currently known about birth order and human capital formation in modern populations. Second, we discuss the mechanisms through which birth order may influence long-run outcomes. Thirdly, we will review what is known from historical and elite samples. Fourth, we will examine family institutions and inheritance. Lastly, we will discuss how this dissertation adds to the existing literature.

2.1 Birth Order and Human Capital Formation

A large body of literature looks at birth-order effects in human capital accumulation. Theoretical frameworks such as the quantity-quality trade-off model developed by Becker and Tomes (1976) emphasized that parental resources (money, time, attention) are finite. Therefore, as family size increases, these investment resources are distributed among a larger number of children. This may reduce average investment per child and disadvantage later-born siblings. This framework is closely related to the resource dilution perspective developed by Blake

(1981), which emphasizes how increases in family size may reduce investments per child.

Modern studies using administrative data and within-family comparisons find a robust "firstborn advantage" or "firstborn premium" in educational attainment and outcomes. Using Norwegian administrative data for cohorts born during the twentieth century, Devereux et al. (2005) show that later-born children complete approximately 0.2 to 0.3 fewer years of schooling. Because comparisons are made within families, these estimates are neither driven by differences in parental background nor socio-economic status across households.

Using U.S. family survey data from the late 20th century, Behrman and Taubman (1986) find that firstborn children obtain, on average, about 0.3 additional years of schooling. Using Canadian survey data from the late 20th century and early 21st century, Kantarevic and Mechoulan (2006) find that second-born children are 3 to 4 percentage points less likely to complete university education than firstborns, reinforcing the existence of birth-order differences in educational outcomes.

Similar results are found in other contexts. Using British household survey data from the early 2000s, Booth and Kee (2009) show that birth order affects educational attainment. They show that firstborn children have higher levels of schooling and academic performance in comparison with their later-born siblings.

Taken together, these findings show that birth order is an important determinant of capital formation and a source of intra-family inequality.

While this literature establishes a robust firstborn advantage in modern populations, it remains unclear whether these patterns extend to pre-industrial contexts characterized by higher fertility, different inheritance norms, and elite selection into education. We extend the birth-order literature to a historical population of scholars and examine whether the modern earlier-born siblings advantage is a historically persistent phenomenon.

2.2 Mechanisms: Parental Investment and Sibling Dynamics

There have been multiple proposals of mechanisms that might explain why birth order matters. One important explanation is differential parental investment. Using U.S. survey data from the early 2000s, Price (2008) shows that "a first-born child receives 20-30 more minutes of quality time each day with his or her parent than a second-born child of the same age from a similar family". These early differences accumulate over time and may have effects on skill formation.

Later research emphasizes strategic parental behavior. Using data from the late 20th century from the U.S., Hotz and Pantano (2015) show that parents respond to early indicators of skills and school performance by strengthening initial advantages rather than compensating weaker outcomes. Thus, relatively minor initial differences in cognition may increase with age, thereby adding to long-standing inequalities by birth-order.

Beyond parental investment mechanisms, there is a broader body of literature that emphasizes the importance of family background on individual outcomes. Studies using Scandinavian data on modern cohorts show that a large share of inequality can be attributed to

factors shared within families, many of which are difficult to observe directly (Björklund and Jäntti, 2020). This perspective reinforces the relevance of within-family comparisons when studying birth-order effects.

Interaction with siblings also plays an important role. Later-born children grow up with less undivided parental attention and in environments where older siblings may act both as role models and as competitors. While psychological mechanisms are not the primary focus of economic models, these approaches are complemented by evidence such as Sulloway (1996), which argues that birth-order differences in non-cognitive traits, including risk-taking and openness to unconventional ideas, may persist over time.

In general, the literature identifies three principal channels through which birth order influences long-run outcomes: resource dilution; dynamic parental investment; and sibling competition.

Most existing studies focus on modern societies with formal schooling. This dissertation explores whether the same birth-order patterns can also be found among scholars from elite families in premodern times.

2.3 Birth Order in Historical and Elite Contexts

While the literature on historical birth-order effects is growing, it remains relatively limited and provides mixed evidence. The nature of fertility, mortality, educational institutions, and labor markets in pre-industrial societies was different from that of modern societies. These differences raise an important question: do the birth-order patterns documented in modern populations generalize to earlier periods?

Using large-scale census data for the United States between 1850 and 1940, Cools et al. (2024) document stable firstborn advantages in occupational and social outcomes over time, despite major demographic and institutional changes. These findings indicate that birth-order effects can persist across periods of structural transformation, although their magnitude remains relatively modest.

Recent evidence from historical administrative data further supports the existence of birth-order effects in pre-industrial settings. Using data from the Netherlands during the 19th and early 20th century, Nuevo-Chiquero et al. (2022) find a negative birth-order effect in occupational outcomes, with later-born individuals systematically attaining lower-ranked occupations than their older siblings. Importantly, they find that birth-order inequalities were already present before.

Recent contributions have further expanded the historical perspective by examining birth-order effects across the demographic transition and over very long time horizons. Using linked Dutch historical records from the 19th century, Holthaus and Nuevo-Chiquero (2025) find no systematic relationship between birth order and survival or longevity once individuals reach adulthood, with only small and context-specific effects in early-life outcomes. This suggests that, in pre-industrial environments characterized by high mortality and limited human

capital accumulation, birth-order differences may have been muted or offset by other family and environmental factors.

Recent evidence using large-scale genealogical data suggests that birth-order effects may in fact be limited in historical contexts. Clark and Cummins (2024) analyze more than 400,000 individuals in England from 1680 to the present day and find a small impact of birth order on social outcomes in the general population. While a firstborn advantage is observed among a small group of elite families in earlier periods, birth order does not significantly affect outcomes for later-born children or across the broader population. These findings highlight that birth-order effects may be highly context-dependent and weaker in historical settings than in modern data. In this sense, focusing on scholars as a selected elite group may provide a setting where birth-order effects are more likely to emerge.

Taken together, these contributions suggest that birth-order effects in historical contexts are more heterogeneous than in modern data and may depend strongly on the institutional and demographic environment.

Likewise, De la Croix and Goñi (2024) separate inherited human capital from nepotism using father-son links among European scholars from the Middle Ages to the Enlightenment. They demonstrate that while the intergenerational transmission of upper-tail human capital was strong, nepotism declined during the Scientific Revolution and Enlightenment. These results indicate that family background mattered in academia, but not all the mechanisms evolved through time as meritocratic pressures rose.

These studies document intergenerational persistence and the evolution of elite reproduction. But they do not systematically consider whether birth order within families influenced scholarly outcomes. It is an open empirical question whether the advantages that firstborns enjoy, as documented in modern data, were already present among historical elites.

A small but growing literature using biographical data on European scholars between the Renaissance and the Industrial Revolution suggests that family background and early-life conditions were strongly correlated with long-run outcomes. Biographies of notable individuals are filled with systematic associations between family structure, longevity, and productivity; early family environments left marks that persisted even among highly selected cohorts (De la Croix and Licandro, 2015; De la Croix et al., 2024).

Another recent research work has looked into the spatial and institutional factors behind elite creativity and intellectual performance. For instance, Serafinelli and Tabellini (2022) demonstrate that the presence of creative elites in cities had persistent positive impacts for those Italian cities during the Renaissance and early modern period in terms of innovation and economic development. Conversely, the emergence of new large scale datasets on relevant historical individuals, like that constructed by Laouenan et al. (2022), allows us to study the long-run rise and formation of European elites from the late Middle Ages to the Industrial Revolution.

This literature emphasises the importance of elite formation for long-run development,

but does not consider intra-family dynamics like birth order. Whether elite trajectories were shaped by within-family hierarchy is directly relevant to this broader literature on the origins of creativity and upper-tail human capital.

Yet much of the existing work does not exploit within-family identification or associate birth order with comprehensive long-run achievement measures, which leaves open if birth-order patterns we observe today were already present among historical elites. Moreover, the mixed and often small magnitude of historical birth-order effects underscores the importance of examining specific elite populations and detailed productivity measures, where intra-family differences may be more likely to emerge.

Taking into account all this literature on historical elites and intergenerational transmission, this dissertation suggests that birth order may represent an additional intra-family dimension associated with elite formation, an aspect of family dynamics that has so far been largely neglected in scholarship.

This dissertation builds on the literature on historical elites and intergenerational transmission by examining the role of birth order within families. By comparing brothers and linking birth order to long-run measures of scholarly achievement, it provides new evidence on whether family hierarchy influenced academic success among historical elites.

2.4 Family Institutions and Inheritance Systems

One important dimension missing from the birth-order literature is that of family institutions. European regions between the sixteenth and nineteenth centuries exhibited different family systems, especially in their inheritance rules and household organization Todd (1990). In primogeniture systems, the eldest son inherited the assets; while in egalitarian systems, the assets were divided equally among the siblings.

These differences in family organization go beyond inheritance rules alone. As emphasized by Baudin et al. (2022), family structures such as nuclear, stem, or complex families shape household decision-making processes, including investments in children and the allocation of resources within families. These institutional differences may therefore interact with birth order in determining individual outcomes.

Recent work indicates that customary institutions had long-term impacts on economic and demographic outcomes. For example, the Customary Atlas of Ancien Régime France covering French customary law before the Revolution from Gay et al. (2024) shows extensive local variation in customary law, including inheritance rules, marriage regulations, and parental authority rights. Such an exercise is useful to exploit spatial variation in family institutions for the identification of causal effects.

New studies using the Customary Atlas find that inheritance rules were closely tied to long-run regional outcomes. The research demonstrates how customary inheritance systems influenced fertility behavior, regional development dynamics, income distribution, poverty and, household formation (Gobbi et al., 2026). Egalitarian inheritance regions were different from

areas with primogeniture, suggesting a spatial variation in inheritance customs. Such family institutions have long-run implications for regional development, inequality, and social norms according to recent economic research (Duranton et al., 2009; Gobbi et al., 2026).

In historical settings where the birth-order effects were likely influenced not only by biological or behavioral factors, but also by institutional context. Inheritance rules can maximize or minimize birth-order advantages, especially among elite families.

While recent research studies the long-run effects of inheritance systems on regional development, nothing is known about how these institutions determined inequality within families. This dissertation fills this gap by investigating whether regional inheritance regimes shaped within-family inequality among scholars, and in particular whether they reinforced or mitigated birth-order differences in long-run intellectual productivity.

2.5 Contribution Relative to the Literature

Instead of being contributive along a single dimension, this dissertation connects across multiple strands of literature. It takes the birth-order literature to a historical sample of European scholars between the thirteenth and eighteenth centuries contributing to debates on elite formation and intergenerational transmission of privilege incorporating regional family institutions into intra-family inequality. This creates a single lens connecting birth order, institutions, and upper-tail human capital in the long-run.

3. Data, Empirical Strategy and Results

This section describes the data sources, the construction of the analytical sample, and the empirical strategy, while also presenting the main descriptive patterns and empirical results. The dataset consists of European scholars and includes family relationships, academic careers, multiple measures of productivity and recognition, biographical traits, and regional inheritance systems. The empirical approach takes advantage of within-family variation between brothers to estimate birth-order effects controlling for unobserved family-level characteristics.

3.1 Data Sources, Sample Construction and Descriptive Evidence

The dataset is constructed using the Repertorium Eruditorum Totius Europae (RETE) database, a secondary prosopographical source that provides biographical information on European scholars and is based on a wide range of historical materials (Curtis et al., 2025). At the individual level, RETE includes information on year of birth and death, place of birth, and detailed records of academic activities and careers.

This secondary source is complemented with information drawn directly from primary historical sources, including published biographical repertories of university professors and academic elites. These repertories provide individual entries and often contain information on family relationships, such as fathers and siblings, which are used to identify and reconstruct

family links between individuals.

When direct information on brothers in the secondary source or in a biographical dictionary is available, we use it. Second, we construct additional brother pairs by locating fathers with multiple sons in the academic records. For each father with at least two sons, all possible pairs of brothers are created.

Brother pairs are treated as unordered pairs and duplicates are removed. Pairs in which both siblings refer to the same individual are excluded. Birth order is determined using their birth year. A pair is classified as *rankable* if the birth year is available for both brothers and the recorded birth years differ across siblings, allowing the unambiguous identification of the older and younger brother. Pairs are classified as *unrankable* when the birth year is missing for at least one sibling or when both brothers share the same recorded birth year, making birth order impossible to establish with certainty. These observations are excluded from the birth-order regressions because the identification of older and younger siblings would otherwise require arbitrary assignment.

The final analytical sample used in the regression analysis consists of 645 rankable pairs of brothers for which birth order can be identified. Unrankable pairs are preserved for descriptive purposes and for analyzing how the composition and data quality of the sample evolve over time, although they are excluded from the empirical birth-order analysis. Appendix Figure 5 presents the distribution of rankable and unrankable brother pairs across centuries. The figure shows that observations with identifiable birth order are available throughout the full historical period covered by the dataset.

Appendix Table 6 compares rankable and unrankable brother pairs. Rankable pairs display substantially higher scholarly visibility and lower rates of zero productivity, suggesting that observations with identifiable birth order are positively selected toward scholars with richer historical records.

In total, approximately 71% of identified brother pairs are rankable, while the remaining observations are classified as unrankable due to missing or identical year of birth.

The summary statistics for the analytical sample are reported in Table 1. Information on the timing of entry into academia is proxied using the first recorded academic activity for each individual. Appendix Figure 6 presents the distribution of age at entry into academia. While most scholars enter in early adulthood, a small number of observations display very early entry ages, which may reflect early university enrollment or measurement imprecision in historical records.

Table 1: Descriptive Statistics

Statistic	Value
Number of brother pairs	645
Mean entry age in academia	32.93
Mean age gap (years)	7.22
Father in academia (share)	38.8%
Titles (older brother)	11.89
Titles (younger brother)	9.79
Older brother more productive (share)	41.6%
Younger brother more productive (share)	31.3%
Share of individuals with no titles	40.2%
Share of pairs with no titles	25.4%

Notes: Descriptive statistics for the sample of rankable brother pairs, i.e. pairs for which birth order can be identified, unless otherwise indicated. Scholarly productivity is measured using the number of titles recorded in VIAF. The mean entry age into academia is computed using all scholars with available information on birth year and first recorded academic activity. Shares are reported as percentages.

There is considerable variation in age gaps between brothers, with an average of 7.2 years. The distribution of age gaps is presented in Appendix Figure 7. The presence of large age differences highlights the importance of controlling for age gaps in the empirical analysis, in order to distinguish birth-order effects from cohort and life-cycle differences.

About 39% of scholars in the sample had a father who was also active in academia. The sample is drawn from the same historical database used in De la Croix and Goñi (2024). We therefore control for paternal academic background in the empirical analysis.

Older brothers appear slightly more productive on average. The average number of titles associated with VIAF is 11.89 for older brothers and 9.79 for younger brothers. However, this difference does not imply that older brothers systematically outperform their siblings. Figure 1 illustrates the distribution of within-family productivity comparisons: older brothers are more productive in 42% of pairs, while younger brothers outperform their older siblings in about 31% of cases, and productivity is equal in the remaining pairs. An additional important feature of the sample is the large share of scholars with no recorded titles: approximately 40% of individuals and 25% of brother pairs have zero observed titles.

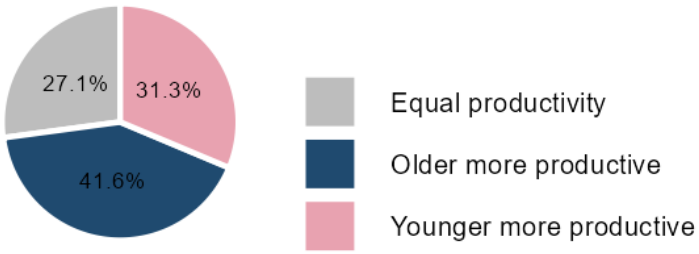


Figure 1: Productivity comparison between older and younger brothers

Notes: The figure shows the distribution of within-family productivity comparisons between older and younger brothers. Shares indicate the proportion of brother pairs in which the older brother has more titles recorded in VIAF, the younger brother has more titles, or both brothers have the same number of titles.

This pattern reflects substantial heterogeneity in within-family productivity differences. While there is a modest average advantage for older brothers, productivity differences vary widely across families and are far from universal. This heterogeneity is particularly visible across historical periods, as shown in Appendix Figures 8 and 9.

Additional descriptive statistics by historical period are presented in Appendix Table 7. The table shows that scholars observed before the sixteenth century tend to exhibit higher average productivity levels and larger age gaps between siblings, although the earlier subsample is substantially smaller.

While the dataset provides a rich and detailed perspective, it reflects a selected sample of scholars who left observable traces in historical records and may therefore over-represent individuals with higher visibility. In addition, bibliographic indicators such as VIAF title counts may partly reflect the long-run accumulation of editions, translations, republications, and archival preservation over time. Although the empirical analysis controls for historical periods, scholars from earlier centuries may have had more opportunity to accumulate observable bibliographic traces in historical records.

This heterogeneity can be illustrated with concrete examples from the dataset. In some cases, differences between siblings are relatively small. For instance, the Bernoulli brothers, well-known figures in the history of mathematics, display very similar levels of scholarly productivity, with Jacob Bernoulli having 63 titles recorded in VIAF and Johann Bernoulli 59. Their prominence in the scientific revolution makes them a natural benchmark for comparing within-family outcomes.

In other cases, however, productivity differences are substantially larger. The Thomasius brothers provide a contrasting example: Christian Thomasius is associated with 71 titles, whereas his younger brother Gottfried Thomasius has only 5. In this case, we did not select the pair of brothers for its historical prominence, but because it exhibits one of the largest productivity gaps observed in the data, making it particularly informative for illustrating the extent of within-family heterogeneity.

At the same time, many of the sibling pairs driving this heterogeneity are less visible

in historical narratives. For example, August Ludwig Schott (1751–1787) and his younger brother Andreas Heinrich Schott (1758–1831) can be clearly ranked by age, and both exhibit measurable scholarly activity (with 4 and 6 recorded titles, respectively), yet neither individual has a significant presence in accessible sources. This kind of pair of brothers is identified through prosopographical data. Notably, both are linked to the same academic environment as their father, further illustrating how such patterns can only be uncovered through structured historical sources.

Taken together, these examples highlight the wide dispersion of within-family outcomes and reinforce the idea that birth-order advantages are not universal, but instead coexist with substantial heterogeneity across families.

Because scholarly productivity is highly skewed and rewards may follow a convex function of talent under superstar-type dynamics (Curtis et al., 2025), the analysis uses the inverse hyperbolic sine transformation of titles to obtain a more concave measure of output differences. Figure 2 shows the resulting distribution of scholarly productivity for older and younger brothers. The two distributions largely overlap, although the distribution for older brothers is slightly shifted to the right, indicating a modest productivity advantage. The concentration of observations near zero reflects the highly skewed nature of historical scholarly output.

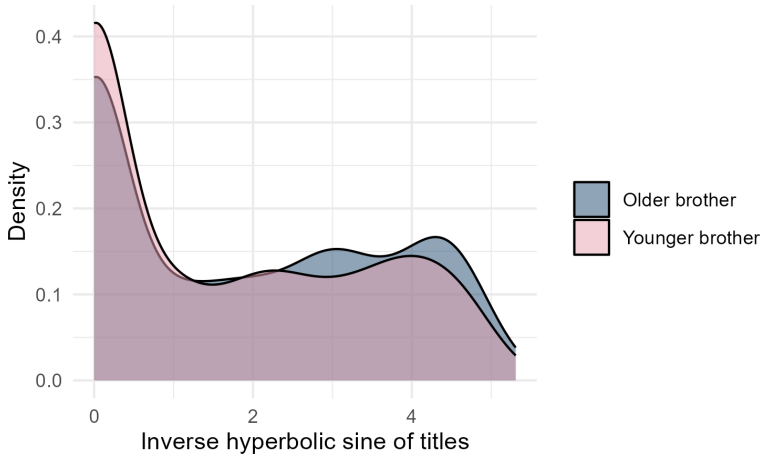


Figure 2: Distribution of scholarly productivity by birth order

Notes: The figure compares the distribution of scholarly productivity between older and younger brothers. Productivity is measured using the inverse hyperbolic sine transformation of the number of titles recorded in VIAF.

3.2 Outcome Variables and Paired Comparisons

The primary outcomes of interest reflect academic productivity and recognition. Productivity is measured using bibliometric indicators derived from the Virtual International Authority File (VIAF), including the number of titles attributed to an individual, as well as the number of publishers, publication countries, coauthors, and works with ISBN identifiers.

Recognition is proxied using Wikipedia-based measures, including the length of articles and the number of languages in which an individual is represented. These measures capture

visibility and perceived importance in the historical record, complementing bibliographic indicators of scholarly output, although they may not fully reflect underlying intellectual ability or informal scholarly contributions not recorded in historical sources.

Additional measures derived from Wikidata include the number of occupations associated with each scholar and the number of recorded academic affiliations. These variables aim to capture broader dimensions of scholarly activity and institutional involvement beyond bibliographic output (Curtis et al., 2025).

As these variables are highly skewed with a large number of zero observations, the analysis uses the inverse hyperbolic sine transformation. Formally, this transformation is defined as: $\text{asinh}(x) = \ln(x + \sqrt{x^2 + 1})$.

The inverse transformation is given by: $\sinh(z) = (e^z - e^{-z})/2$. This allows changes in asinh units to be translated back into the original scale of titles.

This transformation behaves similarly to the logarithmic transformation for large values of x, while remaining well-defined at zero. Unlike a standard logarithmic specification, percentage interpretations of differences in asinh-transformed variables depend on the baseline level of the variable (Aihounton and Henningsen, 2021). To facilitate the interpretation of coefficients in asinh-transformed units, implied percentage differences are evaluated at representative productivity levels using:

$$\%\Delta y = 100 \times \left(\frac{\sinh\left(\text{asinh}(y_0) + \beta\right)}{y_0} - 1 \right),$$

where y_0 denotes the baseline productivity level and β the estimated difference in asinh-transformed outcomes. This allows approximate percentage interpretations while accounting for the non-linear behavior of the transformation near zero.

To ensure that the results are not driven by the choice of functional form, additional robustness checks reported later in the paper replicate the main specifications using both logarithmic and level specifications of productivity differences.

By assigning outcomes independently to the older and younger brothers for rankable pairs, we conduct direct within-family comparisons. Table 2 presents paired comparisons of older and younger brothers across all productivity and recognition measures.

Older brothers tend to perform slightly better than their younger siblings on several dimensions of scholarly productivity. On average, older brothers hold 2.10 more titles in VIAF than younger brothers, a difference that is statistically significant at conventional levels.

Evaluated at the mean productivity level of younger brothers ($y_0 = 9.79$ titles), the asinh difference of 0.245 implies an approximate productivity advantage of 27.5% for older brothers. We used the formula presented previously to compute this percentage: $100 \times \left(\frac{\sinh(\text{asinh}(9.79) + 0.245)}{9.79} - 1 \right) \approx 27.5\%$.

Because the inverse hyperbolic sine transformation is non-linear near zero, this interpretation should be viewed as an approximation rather than as a constant semi-elasticity.

This pattern also appears when considering alternative measures of scholarly prominence. Older siblings tend to have longer Wikipedia pages, more language versions, more publishers, and more coauthors. They are also more likely to have at least one title recorded in VIAF, indicating a higher probability of entry into scholarly production.

However, the size of these differences varies across indicators and is not statistically significant for all outcomes. This suggests that while there is a consistent tendency for older brothers to outperform younger siblings, the magnitude of the effect remains moderate.

Appendix Table 8 reports paired comparisons separately before and after the sixteenth century. The older-brother productivity advantage appears larger in the earlier subsample, although estimates for the pre-sixteenth-century period are based on a considerably smaller number of observations. In particular, the mean asinh productivity difference reaches 1.046 before the sixteenth century, compared with 0.202 in later periods.

Overall, the evidence from paired comparisons points to a modest but systematic older-brother advantage in scholarly productivity and visibility. At the same time, the heterogeneity observed across families indicates that birth order is only one of several factors shaping academic outcomes.

This finding is consistent with the literature on modern populations, which documents an older-brother advantage in educational attainment and long-run outcomes (Devereux et al., 2005; Behrman and Taubman, 1986). However, the magnitude of the effect observed in this historical sample is relatively modest, in line with recent historical evidence suggesting weaker or more context-dependent birth-order effects (Clark and Cummins, 2024).

Table 2: Paired Differences Between Older and Younger Brothers

Measure	Mean Diff.	p-value	N
Titles (VIAF)	2.096	0.0076	645
Titles (VIAF, asinh)	0.245	0.0007	645
Longest Wikipedia page (chars)	409.788	0.0453	645
Longest Wikipedia page (asinh)	0.182	0.0228	645
Has at least one title	0.056	0.0089	645
Number of Wikipedia languages	0.659	0.0215	645
Publication countries (VIAF)	0.372	0.0140	645
Publishers (VIAF)	0.667	0.0043	645
Coauthors (VIAF)	1.056	0.0001	645
Publications with ISBN	0.381	0.1158	645
Publications before 1900	21.667	0.0013	645
Publications after 1900	1.932	0.1904	645
Academic affiliations (Wikidata)	0.130	0.0626	645
Number of occupations (Wikidata)	0.171	0.0328	645
Number of positions	-0.033	0.5849	645
Longevity (years)	-0.180	0.8297	579

Notes: Paired t-tests comparing older and younger brothers across productivity and recognition measures. Mean differences are calculated as older minus younger. Variables transformed using the inverse hyperbolic sine transformation are discussed in the main text.

3.3 Extensive and Intensive Margins of Productivity

To better understand the mechanisms underlying birth-order differences in scholarly productivity, we decompose academic output into extensive and intensive margins. This distinction allows us to assess whether birth order primarily affects entry into scholarly activity or productivity conditional on participation.

The extensive margin captures whether a scholar has at least one publication recorded, while the intensive margin measures productivity conditional on having published. The extensive margin is estimated using a logit specification, whereas the intensive margin is estimated using OLS with an inverse hyperbolic sine transformation of titles. Both specifications control for age gaps between siblings and whether the father was active in academia, with standard errors clustered at the family level.

Table 3 presents the regression results. The estimates show that older brothers are significantly more likely to have at least one publication recorded. The coefficient on the indicator for being the older sibling is positive and statistically significant, indicating that older siblings are significantly more likely to enter scholarly production. The estimated coefficient of 0.238 in the logit specification corresponds to an odds ratio of approximately $e^{0.238} \approx 1.27$, implying that older brothers have approximately 27% higher odds of having at least one recorded publication than their younger siblings. In other words, holding age gap and family background constant, older brothers are substantially more likely to appear in the historical record with at least one identified publication.

Although the coefficient on the older-sibling indicator remains positive in the intensive-margin specification, it is statistically insignificant, suggesting that conditional productivity differences are limited once scholars enter academic production.

Taken together, these results indicate that birth-order effects operate primarily through the extensive margin. Older siblings are more likely to become active scholars, while conditional productivity differences remain limited. This pattern is consistent with theories of parental investment, which emphasize differential access to opportunities rather than performance among those who participate (Becker and Tomes, 1976; Hotz and Pantano, 2015). In historical contexts in which access to academic careers required substantial family investment and social connections, older siblings may have benefited from preferential allocation of resources, such as education or access to networks, increasing their likelihood of entering academia. Once these barriers are overcome, however, long-run productivity appears to depend more on individual ability and institutional opportunities than on birth order, although this interpretation relies on observed publication records and cannot capture unrecorded scholarly activity.

Table 3: Extensive and Intensive Margins of Scholarly Productivity

	Extensive (Logit) *Pr(Title > 0)*	**Intensive (OLS)** *Asinh titles*
Constant	0.051	2.892***
	(0.129)	(0.101)
Older sibling	0.238***	0.141
	(0.091)	(0.089)
Age gap	-0.002	-0.013
	(0.012)	(0.009)
Father in academia	0.654***	0.090
	(0.176)	(0.112)
Clustered SE (family)	Yes	Yes
Observations	1290	772
R^2		0.008
Adj. R^2		0.004

Notes: Column (1) reports logit estimates for the probability of having at least one title recorded in VIAF. Column (2) reports OLS estimates conditional on positive titles using the inverse hyperbolic sine transformation of titles. Standard errors are clustered at the family level. *Significance levels:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.4 Institutional Environment and Inheritance Systems

To account for institutional variation across regions, the analysis incorporates information on inheritance systems following the classification developed by Todd (1990), with adaptations for France based on Gay et al. (2024). Regions are categorized according to their predominant inheritance regime, distinguishing between egalitarian and non-egalitarian systems.

Inheritance regimes are assigned to scholars based on their birthplace using a geographic mapping at the NUTS3 level, although this regional assignment may mask within-region heterogeneity in actual family practices. For brother pairs with known birth order, the inheritance regime is associated with the birthplace of the older brother. Because the data only contain scholars observed in RETE, the older observed brother may not always coincide with the biological firstborn of the family, implying that the analysis captures birth-order differences among observed scholarly siblings rather than strict primogeniture status.

Figure 3 provides a geographic visualization of inheritance systems alongside the birthplaces of scholars in the dataset. Shaded areas represent Todd's typology of inheritance regimes, while circles indicate the number of scholars born in each location. Scholars in the dataset are distributed across a wide range of European regions, spanning diverse institutional environments. Among observations with classified inheritance systems, approximately 47% correspond to non-egalitarian regimes and 30% to egalitarian ones. A non-negligible share of observations remains unclassified because some regions cannot be matched precisely to Todd's typology. These proportions are reported in more detail in Appendix Table 9, which also shows that approximately 23% of observations remain unclassified because some regions are not assigned to either egalitarian or non-egalitarian inheritance systems in Todd's typology.

This variation across space provides the basis for examining whether inheritance institutions are associated with the within-family productivity differences.

To capture the temporal dimension of this variation, Figure 4 shows the share of brother pairs born in regions characterized by non-egalitarian inheritance systems by century.

Figure 4 indicates that non-egalitarian inheritance systems remain prevalent across the sample, although their relative importance varies over time. This spatial and temporal variation provides the empirical foundation for examining whether inheritance institutions are associated with within-family productivity differences across historical periods.

Importantly, Figure 4 also shows that non-egalitarian inheritance systems are almost absent in the thirteenth and fourteenth centuries of the sample. For this reason, some supplementary specifications reported in the Appendix distinguish between observations before and after the sixteenth century, in order to account for the limited variation in inheritance systems during the earlier centuries of the sample.

In principle, inheritance institutions may influence parental investment decisions. In non-egalitarian systems, such as primogeniture, family resources and opportunities may have been concentrated toward older sons, potentially reinforcing birth-order advantages. In contrast, egalitarian systems may distribute resources more evenly across siblings, reducing such differences.

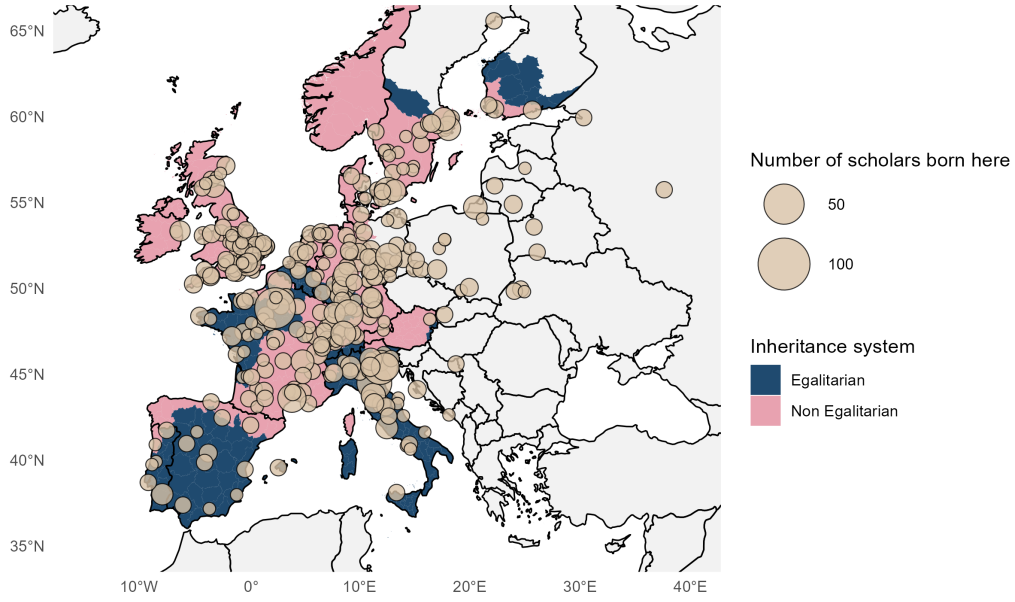


Figure 3: Birthplaces of Scholars and Inheritance Systems in Europe

Notes: The map displays the birthplaces of scholars belonging to the analytical sample of rankable brother pairs with identifiable birth locations, together with the classification of inheritance systems across European regions. Inheritance regimes are assigned at the NUTS3 level following Todd's classification. The figure provides a descriptive illustration of the geographical distribution of rankable scholarly brother pairs and inheritance systems across Europe. Circles indicate the number of scholars born in each location.

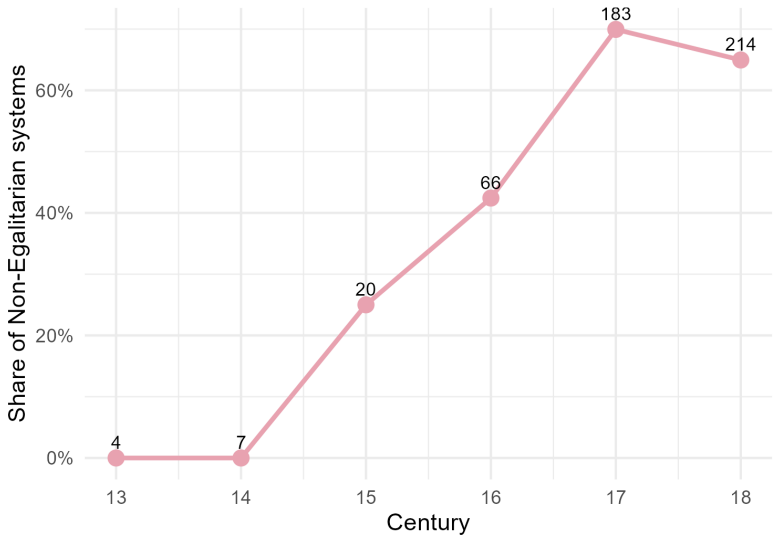


Figure 4: Share of Brother Pairs from Non-Egalitarian Inheritance Regions by Century

Notes: The figure shows the share of rankable brother pairs from non-egalitarian inheritance regions by century. Inheritance regimes are assigned based on the birthplace of the older brother, following Todd's classification. Numbers above the points indicate the total number of brother pairs observed in each century.

3.5 Regression Analysis

To quantify birth-order differences in scholarly productivity, we estimate regression models where the dependent variable is the inverse hyperbolic sine transformation of the difference in title counts between older and younger brothers. Positive values indicate higher productivity for the older sibling. Because the empirical specification relies on comparisons between older and younger siblings, brother pairs with identical recorded birth years are excluded from the regression analysis.

The baseline empirical specification is estimated using ordinary least squares and can be written as:

$$\begin{aligned}\Delta Productivity_i = \beta_0 &+ \beta_1 AgeGap_i + \beta_2 FatherAcademia_i \\ &+ \beta_3 DifferentInstitution_i + \beta_4 NonEgalitarian_i \\ &+ \beta_5 UnknownInheritance_i + \varepsilon_i\end{aligned}$$

where $\Delta Productivity_i = \text{asinh}(Titles_i^{Older}) - \text{asinh}(Titles_i^{Younger})$. This specification measures within-family differences in scholarly productivity in transformed units while remaining well-defined in the presence of zero publication counts.

$AgeGap_i$ is a continuous variable measuring the age difference in years between siblings. $FatherAcademia_i$ is a binary indicator equal to one if the father was active in academia. $DifferentInstitution_i$ is a binary indicator equal to one if the two brothers are associated with different academic institutions. $NonEgalitarian_i$ and $UnknownInheritance_i$ are binary indicators capturing the inheritance regime of the birthplace region of the older brother, with egalitarian inheritance systems used as the omitted reference category.

In this specification, β_1 captures how productivity differences vary with sibling age gaps, β_2 captures differences associated with paternal academic background, β_3 measures the association between institutional divergence and productivity gaps, while β_4 and β_5 capture differences associated with inheritance systems relative to egalitarian regions.

The constant term therefore represents the expected productivity difference for brother pairs from egalitarian inheritance regions, sharing the same institution, without a father in academia, and with an age gap equal to zero. Since observations with exactly zero age gaps are largely absent from the sample, the constant should not be interpreted literally but rather as a baseline reference category for the regression.

The preferred specification additionally includes century indicator variables in order to account for broad historical differences in scholarly production, intellectual institutions, and the preservation of historical records across periods. The specification becomes:

$$\begin{aligned}\Delta Productivity_i = \beta_0 &+ \beta_1 \textit{AgeGap}_i + \beta_2 \textit{FatherAcademia}_i \\ &+ \beta_3 \textit{DifferentInstitution}_i + \beta_4 \textit{NonEgalitarian}_i \\ &+ \beta_5 \textit{UnknownInheritance}_i + \sum_{c=13}^{17} \gamma_c Century_{ic} + \varepsilon_i\end{aligned}$$

where $Century_{ic}$ is an indicator variable equal to one if brother pair i belongs to century c, and zero otherwise, for centuries thirteen through seventeen. The eighteenth century is used as the omitted reference category. Each γ_c therefore captures the difference in the older-younger productivity gap relative to eighteenth-century brother pairs, holding the other covariates constant.

Table 4 reports the results. Across specifications, the estimates suggest modest but persistent productivity differences favoring older brothers, although the mechanisms underlying these differences vary across historical periods. In baseline specifications, larger age gaps between siblings are associated with larger productivity differences. The positive coefficient on age differences suggests that, holding other observed characteristics constant, brother pairs with larger age gaps tend to display somewhat larger productivity differences in favor of the older sibling, although this relationship becomes statistically weaker once century indicators are introduced.

Family background does not appear to play a statistically significant equalizing role. The coefficient on father in academia is negative but not statistically significant, suggesting no robust evidence that shared access to academic networks reduces within-family productivity differences.

In contrast, differences in institutional affiliation emerge as a more robust determinant of within-family productivity differences. Brother pairs affiliated with different academic institutions tend to display larger productivity gaps across specifications, particularly after the sixteenth century. Across the main specifications, the coefficient on different institutions is

0.485. Evaluated at average productivity levels, this implies a productivity gap approximately 61.4% larger for brother pairs affiliated with different institutions:

$$100 \times \left(\frac{\sinh\left(\operatorname{asinh}(9.79) + 0.485\right)}{9.79} - 1 \right) \approx 61.4\%.$$

Holding other observed characteristics constant, brother pairs affiliated with different institutions display productivity gaps approximately 61.4% larger on average than pairs sharing the same institutional environment.

This suggests that institutional environments may play an important role in shaping divergent scholarly trajectories and that shared institutional environments may partially attenuate within-family productivity differences.

Additional regressions reported in Table 5 reveal substantial heterogeneity across historical periods and suggest that productivity differences between brothers were driven by different mechanisms before and after the sixteenth century.

Before the sixteenth century, larger age gaps between siblings were robustly associated with larger productivity differences across specifications, even after controlling for century fixed effects, although this subsample remains relatively small. These early-period estimates should be interpreted cautiously. The pre-sixteenth-century subsample contains a relatively small number of observations and likely over-represents highly visible scholarly families for whom detailed historical records survive. In addition, institutional and inheritance-system variation is substantially more limited during the earlier centuries of the sample, making it more difficult to disentangle the effects of age gaps from broader historical and selection effects. One possible interpretation is that age differences reflected stronger cohort, seniority, and life-cycle advantages in medieval and early Renaissance academic environments, where access to scholarly careers remained relatively concentrated.

By contrast, after the sixteenth century, the age-gap effect becomes statistically weak, while differences in institutional affiliation emerge as a much more robust determinant of productivity divergence between brothers. This pattern is consistent with the growing differentiation and expansion of European universities and intellectual networks after the early modern period. As academic institutions became increasingly specialized and geographically dispersed, institutional environments appear to have played a more important role in shaping divergent scholarly trajectories within the same family.

Additional specifications reported in Appendix Table 10 examine whether the relationship between inheritance systems and productivity differences changes after the sixteenth century using the full sample. The interaction terms suggest that the relationship between inheritance systems and productivity differences remains insignificant.

These findings suggest that while inheritance systems influenced family structures and economic incentives in historical Europe, their direct effect on productivity gaps was limited. Academic careers likely depended on other factors, including access to institutions

and intellectual networks. This result is in line with recent evidence pointing towards a rather restricted role of family institutions for within-family inequality (Gobbi et al., 2026).

In summary, we do not find evidence that inheritance systems generally have a direct impact on within-family productivity differences. While inheritance regimes influenced family structures and economic incentives in historical Europe, the empirical results suggest that institutional environments and historical context played a more important role in explaining productivity divergence between siblings.

These results contribute to the mixed evidence in the literature. While modern studies find consistent advantages for earlier-born siblings, recent historical evidence points to smaller or less systematic effects (Clark and Cummins, 2024). Our findings suggest that such effects may emerge more clearly within elite populations, such as scholars.

Table 4: Determinants of the Birth-Order Productivity Gap

	(1)	(2)	(3)	(4)
	Asinh difference in titles			
Constant	0.083	0.023	0.005	-0.162
	(0.109)	(0.126)	(0.177)	(0.197)
Age gap	0.023**	0.022**	0.022**	0.019*
	(0.010)	(0.010)	(0.010)	(0.011)
Father in academia		-0.184	-0.186	-0.163
		(0.149)	(0.148)	(0.150)
Different institution		0.440**	0.440**	0.485***
		(0.181)	(0.181)	(0.180)
Non-egalitarian inheritance			0.031	0.135
			(0.169)	(0.175)
Unknown inheritance system			0.019	0.130
			(0.198)	(0.205)
13th century				0.099
				(0.223)
14th century				1.764***
				(0.552)
15th century				0.863**
				(0.345)
16th century				0.025
				(0.226)
17th century				0.099
				(0.161)
Century controls	No	No	No	Yes
Clustered SE (family)	Yes	Yes	Yes	Yes
Observations	645	645	645	645
R^2	0.005	0.020	0.020	0.035
Adj. R^2	0.004	0.015	0.012	0.020

Notes: The dependent variable is the inverse hyperbolic sine (asinh) difference in titles between older and younger brothers. Positive values indicate higher productivity for the older sibling. Standard errors clustered at the family level are reported in parentheses. Column (4) includes century indicator variables, with the 18th century used as the omitted reference category.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 5: Birth-Order Productivity Differences Across Historical Periods

	(1) *Before 16th century*	(2) *16th century onward*
	Asinh difference in titles	
Constant	-0.207	-0.147
	(0.689)	(0.200)
Non-egalitarian inheritance	-0.719	0.164
	(0.584)	(0.180)
Inheritance unknown	-1.730	0.175
	(1.724)	(0.207)
Age gap	0.110***	0.013
	(0.031)	(0.011)
Father in academia	0.415	-0.160
	(0.771)	(0.154)
Different institution	0.995	0.474**
	(0.643)	(0.184)
Century 13	-1.595***	—
	(0.584)	—
Century 14	1.202	—
	(0.744)	—
Century 16	—	0.032
	—	(0.226)
Century 17	—	0.102
	—	(0.161)
Century controls	Yes	Yes
Clustered SE (family)	Yes	Yes
Observations	33	612
R^2	0.387	0.020
Adj. R^2	0.215	0.008

Notes: The dependent variable is the inverse hyperbolic sine (asinh) of the difference in titles between older and younger brothers. Column (1) reports estimates for scholars born before the sixteenth century, while Column (2) reports estimates for scholars born from the sixteenth century onward. All specifications include century controls. The omitted century category is the 15th century in Column (1) and the 18th century in Column (2). Standard errors clustered at the family level are reported in parentheses.
Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

3.6 Robustness and Final Interpretation

Several different specifications and sample restrictions are applied in order to test the robustness of the birth-order results.

First, we limit the sample to brother pairs with an age difference of no more than twenty years (Appendix Table 11, Column 1). This addresses the concern that very large age differences between siblings may have more to do with differences in generational context than actual sibling comparisons. The estimated coefficients remain broadly similar in magnitude to the baseline results, implying that the primary findings are usually not driven by excessive age variations.

Second, we limit the sample to families with two sons (Appendix Table 11, Column 2).

This specification removes cases in which larger families generate multiple brother pairs and therefore focuses on simpler sibling structures. Despite the smaller sample size, the estimated coefficients remain broadly consistent with the baseline specification.

Third, we repeat the analysis with alternative measures of scholarly visibility (Appendix Table 12). Furthermore, we look at the length of the scholar's Wikipedia page, the number of language versions that exist on Wikipedia, and the number of publishers associated with their works. These variables capture broader dimensions of scholarly visibility and intellectual prominence beyond publication counts.

Fourth, we replicate the main specifications using alternative functional forms for productivity differences. In addition to the baseline inverse hyperbolic sine transformation, the regressions are re-estimated using logarithmic and level specifications of title differences (Appendix Tables 13 and 14). The logarithmic specification is implemented as $\log(x + \varepsilon)$ with $\varepsilon = 0.1$ in order to accommodate zero observations while minimizing distortions for positive productivity values close to zero.

The results obtained using the logarithmic specification are quantitatively very similar to those based on the inverse hyperbolic sine transformation. In the preferred logarithmic specification, the coefficient on different institutions is approximately 0.58. Under the logarithmic specification, percentage differences are interpreted using: $100 \times \left(e^{\beta} - 1\right)$. Applying this transformation gives: $100 \times (e^{0.58} - 1) \approx 78.6\%$.

In the preferred logarithmic specification, holding other observed characteristics constant, brother pairs affiliated with different institutions display productivity gaps approximately 78.6% larger on average than pairs sharing the same institutional environment.

Although this interpretation depends on the choice of the adjustment parameter ε used to accommodate zero observations, the estimated magnitudes remain broadly consistent with the baseline asinh and level specifications.

In particular, the positive association between differences in institutional affiliation and productivity divergence between siblings remains stable across specifications, suggesting that the main findings are not driven by the specific treatment of highly skewed productivity measures.

The level specifications lead to the same qualitative conclusions. In these regressions, coefficients are interpreted directly in terms of differences in the number of recorded titles between older and younger brothers. For example, the coefficient on different institutions in the preferred specification implies that, holding other observed characteristics constant, brother pairs affiliated with different institutions display productivity gaps approximately five titles larger on average. Although the level specification is more sensitive to extreme observations due to the highly skewed distribution of scholarly output, the overall pattern of results remains consistent with the baseline asinh specifications.

The magnitude of the estimates is also broadly comparable across functional forms. Evaluated at the mean productivity level of younger brothers ($y_0 = 9.79$ titles), the preferred

asinh specification implies a productivity gap of approximately 61%, corresponding to roughly six additional recorded titles at the mean productivity level. This magnitude is close to the estimates obtained in the level specifications, reinforcing the overall consistency of the results across alternative specifications.

Additional period-specific specifications for the logarithmic and level regressions are reported in Appendix Tables 15 and 16. These regressions estimate the models separately before and after the sixteenth century in order to evaluate whether the historical heterogeneity identified in the baseline specifications remains robust across alternative functional forms.

The period-specific logarithmic specifications remain closely aligned with the baseline findings. Before the sixteenth century, larger age gaps between siblings remained strongly associated with larger productivity differences, even after controlling for century fixed effects, whereas after the sixteenth century, differences in institutional affiliation emerged as the more robust correlate of within-family productivity divergence.

Some differences nevertheless emerge in the pre-sixteenth-century level specifications. In these regressions, non-egalitarian inheritance systems and differences in institutional affiliation become statistically significant, while the age-gap effect remains positive and statistically significant. Given the small size of the early-period subsample and the sensitivity of level specifications to extreme observations, these estimates likely reflect the influence of a limited number of highly productive scholarly families and the greater impact of outliers in untransformed productivity measures.

Finally, Appendix Table 17 introduces additional country controls in order to account for persistent cross-country differences in institutional environments, economic conditions, and historical academic traditions. The distribution of observations across countries is highly uneven, with a small number of countries accounting for most of the sample. Germany contains 116 observations, followed by the United Kingdom (111), France (104), and Italy (96), while several remaining countries contain only a limited number of observations. Once country controls are included, the main coefficients remain broadly consistent with the baseline specifications, although some estimates become less precisely estimated. Overall, the results suggest that the observed productivity differences are not solely driven by broad country-level heterogeneity, while also indicating that part of the variation may reflect wider national institutional and academic environments.

The robustness exercises considered together suggest that the main findings are not coming from specific sample choices or from the particular measure of scholarly productivity employed. Though the magnitude of the coefficients differs between specifications, the overall pattern remains broadly consistent with a modest birth-order effect in scholarly productivity, although the estimated magnitudes vary across historical periods and empirical specifications.

Overall, the results provide evidence of a modest birth-order effect in scholarly productivity among historical European scholars. This effect appears to operate primarily through differences in access to scholarly activity rather than through conditional productivity

differences among active scholars. The broad consistency of the results across inverse hyperbolic sine, logarithmic, and level specifications further suggests that the main conclusions are not driven by the particular transformation used to measure scholarly productivity.

The findings also indicate substantial historical heterogeneity. Before the sixteenth century, productivity differences between brothers appear more closely associated with sibling age gaps, whereas after the sixteenth century, institutional divergence becomes more strongly associated with within-family productivity differences. This suggests that the increasing differentiation of European academic institutions may have amplified divergent scholarly trajectories within the same family.

The results suggest that institutional environments are more consistently associated with productivity differences than inheritance systems.

4. Conclusion

This dissertation studied the relationship between birth order and academic productivity among historical European scholars. Specifically, using a new dataset of 645 pairs of brothers, the analysis examined whether older brothers showed higher scholarly productivity than their younger siblings and whether these patterns varied across historical periods and institutional environments.

The analysis provides evidence of a slight birth-order difference in scholarly productivity. The estimates indicate a modest but consistent productivity advantage for older brothers across several indicators of scholarly visibility and academic output. Crucially, birth-order differences appear to operate primarily through the extensive margin of scholarly activity: older siblings are more likely to enter scholarly production, while, conditional on publishing, productivity differences between brothers become substantially smaller and statistically insignificant. The results also reveal substantial historical heterogeneity. Before the sixteenth century, productivity differences between brothers appear more closely associated with sibling age gaps, whereas after the sixteenth century, differences in institutional affiliation become more strongly associated with productivity divergence within families. Because the analysis relies on observed scholarly siblings rather than complete family genealogies, the results should be interpreted as evidence on birth-order differences among observed academic elites rather than as direct estimates of strict primogeniture effects. At the same time, the results should be interpreted in light of the selected nature of historical scholarly records and the limited availability of detailed family information for earlier periods.

These results contribute to the existing literature. First, they extend the birth-order literature to a historical population of European intellectual elites, providing evidence that within-family productivity differences associated with birth order were already observable among pre-industrial academic elites. Second, focus on elite formation and intergenerational transmission by showing how sibling hierarchy, age structure, and institutional environments

influenced scholarly trajectories. Finally, the results indicate that institutional and historical contexts were more relevant than inheritance systems in determining productivity differences among scholarly siblings, mainly through differences in institutional affiliation. These findings are also consistent with recent historical evidence suggesting that birth-order effects are limited in the general population but may nevertheless emerge more clearly within selected elite populations such as scholars (Clark and Cummins, 2024).

Future research might extend the analysis to larger sibling groups, allowing for direct comparisons between firstborn, middle-born, and later-born children (that is, both secondborn siblings as well as those born after them in such a group). It can also focus on studying the mechanisms underlying these patterns more directly, including institutional mobility, academic networks, and intergenerational transmission of scholarly capital. Extending the analysis to other elite and non-elite subgroups might clarify the extent to which birth-order dynamics shaped long-run human-capital formation and social mobility in historical societies.

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Appendix

A.1 Rankable and Unrankable Brother Pairs

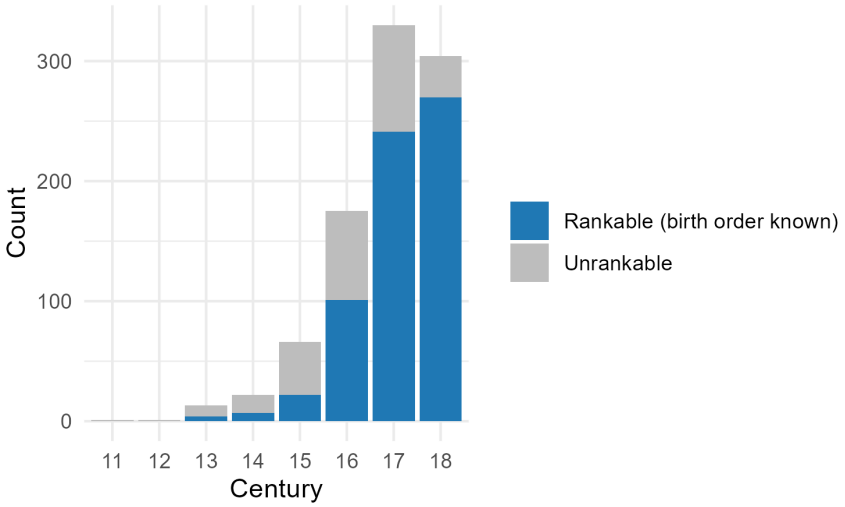


Figure 5: Rankable and Unrankable Brother Pairs Across Time

Notes: The figure shows the distribution of rankable and unrankable brother pairs across centuries. Rankable pairs are those for which birth order can be identified using year-of-birth information.

Table 6: Comparison Between Rankable and Unrankable Brother Pairs

	Rankable	**Unrankable**
Brother pairs	645	268
Mean age gap	7.22	–
Mean entry age	29.43	27.06
Father in academia (share)	0.388	0.410
Mean titles	10.84	1.87
Share individuals with no titles	0.402	0.816
Share pairs with no titles	0.254	0.693

Notes: The table compares rankable and unrankable brother pairs. Rankable pairs are those for which birth order can be identified using year-of-birth information.

A.2 Entry into Academia and Age Gaps

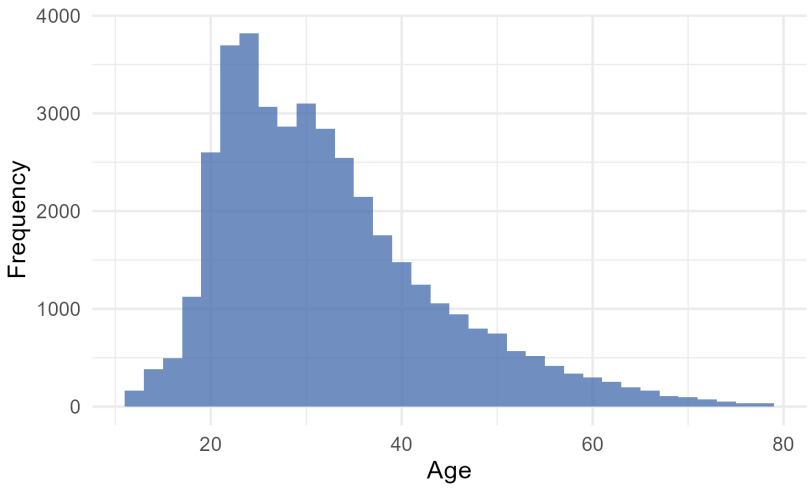


Figure 6: Distribution of Age at Entry into Academia

Notes: The figure presents the distribution of age at first recorded academic activity for scholars with available information on both birth year and academic activity. Age at entry is computed as the difference between the first recorded academic year and the year of birth.

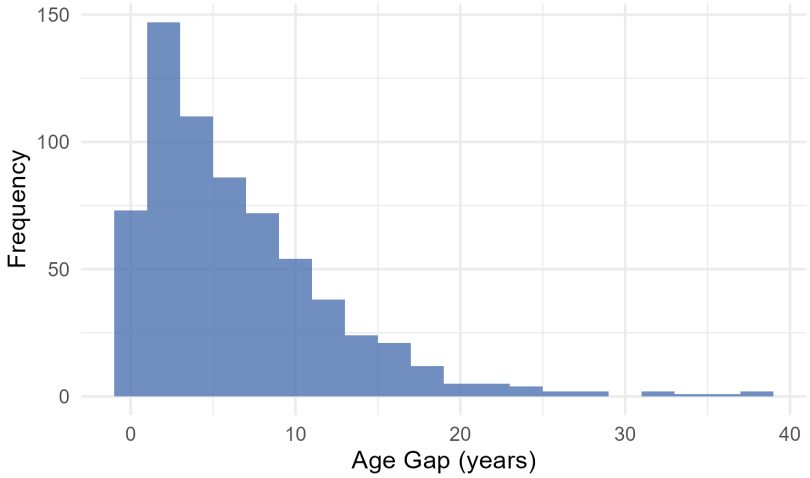


Figure 7: Distribution of Age Gaps Between Brothers

Notes: The histogram displays the distribution of age differences between older and younger brothers in rankable pairs.

A.3 Historical Productivity Patterns

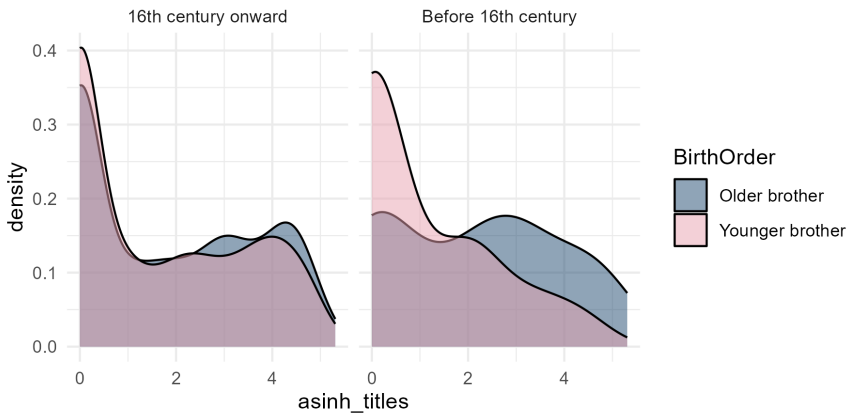


Figure 8: Distribution of Scholarly Productivity Across Historical Periods

Notes: The figure compares the distribution of asinh-transformed scholarly productivity before and after the sixteenth century.

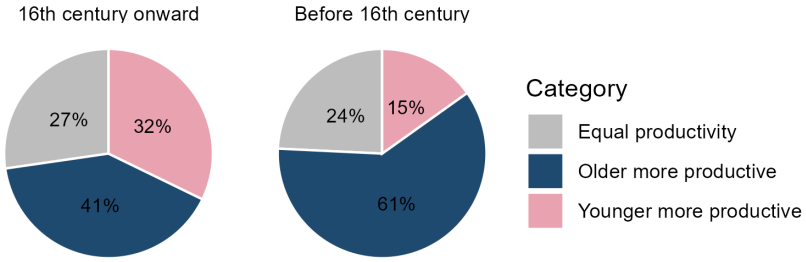


Figure 9: Birth-Order Productivity Comparisons by Historical Period

Notes: The figure compares the proportion of pairs in which older brothers, younger brothers, or neither sibling exhibit higher productivity across historical periods.

Table 7: Birth-Order Differences Before and After the 16th Century

	Before 16th century	**16th century onward**
Brother pairs	33	612
Mean age gap	10.18	7.06
Mean titles (older brothers)	13.94	11.78
Mean titles (younger brothers)	4.45	10.08
Older brother more productive (share)	0.606	0.405
Pairs with no titles (share)	0.242	0.255
Father in academia (share)	0.303	0.392

Notes: Descriptive statistics for the sample of rankable brother pairs, i.e. pairs for which birth order can be identified, reported separately for observations before and after the sixteenth century. Scholarly productivity is measured using the number of titles recorded in VIAF. Shares are reported as proportions.

Table 8: Paired Comparisons by Historical Period

Period	Mean Difference	p-value	N
Before 16th century	1.046***	0.0007	33
16th century onward	0.202***	0.0063	612

Notes: The table reports paired comparisons of the inverse hyperbolic sine (asinh) difference in titles between older and younger brothers across broad historical periods. Mean differences are calculated as older minus younger. *Significance levels:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.4 Inheritance Systems and Historical Composition

Table 9: Inheritance Systems in the Sample

Inheritance system	Count	Share
Egalitarian	194	0.301
Non-egalitarian	300	0.465
Unknown	151	0.234

Notes: Inheritance systems are assigned using Todd's classification at the NUTS3 level. Shares are computed over the full analytical sample.

A.5 Additional Institutional Interaction Specifications

Table 10: Inheritance Systems and Birth-Order Productivity After the Sixteenth Century

	(1) *Asinh difference in titles*
Intercept	0.998***
	(0.305)
Non-egalitarian inheritance	-0.362
	(0.631)
Inheritance unknown	-1.716
	(1.258)
Post-16th century	-1.134***
	(0.314)
Age gap	0.019*
	(0.010)
Father in academia	-0.178
	(0.149)
Different institution	0.472***
	(0.180)
Non-egalitarian × Post-16th	0.533
	(0.655)
Unknown inheritance × Post-16th	1.897
	(1.276)
Clustered SE (family)	Yes
Observations	645
R^2	0.034
Adj. R^2	0.022

Notes: The dependent variable is the inverse hyperbolic sine (asinh) of the difference in titles between older and younger brothers. The specification interacts inheritance systems with an indicator for observations from the sixteenth century onward. The split is motivated by the limited variation in inheritance-system classifications observed before the sixteenth century. Standard errors are clustered at the family level. *Significance levels:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

A.6 Robustness Checks

Table 11: Alternative Sample Specifications

	(1) *Age gap* ≤ 20	(2) *Two-son families only*
	Asinh difference in titles	
Constant	-0.175	-0.151
	(0.209)	(0.223)
Non-egalitarian inheritance	0.153	0.047
	(0.179)	(0.202)
Inheritance unknown	0.167	0.020
	(0.211)	(0.240)
Age gap	0.022	0.014
	(0.016)	(0.013)
Father in academia	-0.168	-0.054
	(0.152)	(0.177)
Different institution	0.478***	0.493**
	(0.182)	(0.202)
13th century	0.080	-0.330
	(0.231)	(0.264)
14th century	1.761***	1.687**
	(0.550)	(0.850)
15th century	0.796**	1.038***
	(0.371)	(0.387)
16th century	-0.023	0.235
	(0.231)	(0.265)
17th century	0.079	0.192
	(0.164)	(0.187)
Century indicators	Yes	Yes
Clustered SE (family)	Yes	Yes
Observations	624	467
R^2	0.033	0.036
Adj. R^2	0.018	0.014

Notes: The dependent variable is the inverse hyperbolic sine (asinh) difference in titles between older and younger brothers. Column (1) restricts the sample to brother pairs with an age gap below or equal to twenty years. Column (2) restricts the sample to families with only two sons. All specifications include century indicator variables, with the eighteenth century used as the omitted reference category. Standard errors are clustered at the family level. Across both robustness specifications, differences in institutional affiliation remain positively associated with larger productivity differences between siblings, while inheritance-system coefficients remain statistically insignificant. *Significance levels:* *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 12: Alternative Measures of Scholarly Visibility

	(1) *Wikipedia length gap*	(2) *Wikipedia languages gap*	(3) *Publishers gap*
Constant	0.077	-0.203	-0.559
	(0.231)	(0.759)	(0.627)
Non-egalitarian inheritance	-0.027	-0.463	0.164
	(0.197)	(0.673)	(0.575)
Inheritance unknown	-0.199	0.053	0.110
	(0.225)	(0.706)	(0.670)
Age gap	0.016	0.050	0.082**
	(0.014)	(0.036)	(0.033)
Father in academia	-0.344**	-0.028	-0.600
	(0.157)	(0.593)	(0.467)
Different institution	0.148	1.885**	1.550***
	(0.187)	(0.818)	(0.562)
13th century	0.812	1.694	-4.146
	(0.343)	(1.483)	(1.392)
14th century	0.139	6.205***	7.036***
	(0.951)	(1.872)	(2.527)
15th century	0.964***	1.438	2.390
	(0.526)	(1.851)	(1.792)
16th century	0.053	0.170	0.271
	(0.241)	(0.787)	(0.497)
17th century	0.237	-0.061	0.305
	(0.172)	(0.770)	(0.310)
Century indicators	Yes	Yes	Yes
Clustered SE (family)	Yes	Yes	Yes
Observations	645	645	645
R^2	0.022	0.026	0.045
Adj. R^2	0.007	0.010	0.030

Notes: Dependent variables are alternative measures of scholarly visibility differences between older and younger brothers. Column (1) uses the inverse hyperbolic sine (asinh) difference in Wikipedia page length. Column (2) uses the difference in the number of Wikipedia language versions. Column (3) uses the difference in the number of publishers associated with VIAF records. All specifications include century indicator variables, with the eighteenth century used as the omitted reference category. Standard errors are clustered at the family level. The results indicate that institutional divergence between siblings is positively associated with broader differences in scholarly visibility, particularly regarding Wikipedia language coverage and publisher networks.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 13: Logarithmic Specification

	(1)	(2)	(3)	(4)
Constant	0.1177	0.0256	0.0034	-0.2192
	(0.1530)	(0.1781)	(0.2419)	(0.2718)
Age gap	0.0302**	0.0297*	0.0296*	0.0244
	(0.0153)	(0.0154)	(0.0155)	(0.0157)
Father in academia		-0.1674	-0.1748	-0.1424
		(0.2036)	(0.2037)	(0.2070)
Different institution		0.5224**	0.5208**	0.5773**
		(0.2427)	(0.2429)	(0.2429)
Non-egalitarian inheritance			0.0640	0.1978
			(0.2328)	(0.2417)
Unknown inheritance system			-0.0163	0.1234
			(0.2692)	(0.2799)
13th century				0.0520
				(0.3011)
14th century				1.8130***
				(0.5829)
15th century				1.4065***
				(0.5039)
16th century				0.0711
				(0.3073)
17th century				0.1507
				(0.2228)
Century indicators	No	No	No	Yes
Clustered SE (family)	Yes	Yes	Yes	Yes
Observations	645	645	645	645
R^2	0.005	0.015	0.015	0.029
Adj. R^2	0.004	0.010	0.007	0.014

Notes: The dependent variable is the logarithmic difference in titles between older and younger brothers, computed as $\log(\text{Titles}_{Older} + \varepsilon) - \log(\text{Titles}_{Younger} + \varepsilon)$, where $\varepsilon = 0.1$. Positive values indicate higher productivity for the older sibling. Standard errors clustered at the family level are reported in parentheses. Column (4) includes century indicator variables, with the eighteenth century used as the omitted reference category.
Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 14: Level Specification

	(1)	(2)	(3)	(4)
Constant	0.6422	-0.1949	-0.4523	-2.3098
	(1.1962)	(1.3309)	(1.9141)	(2.1160)
Age gap	0.2013*	0.1964*	0.1980*	0.1909*
	(0.1028)	(0.1012)	(0.1017)	(0.1030)
Father in academia		-1.4100	-1.3192	-1.0555
		(1.6792)	(1.6453)	(1.6620)
Different institution		4.5978**	4.6302**	5.1599**
		(2.0256)	(2.0235)	(2.0156)
Non-egalitarian inheritance			-0.2075	0.9186
			(1.8637)	(1.9094)
Unknown inheritance system			1.2691	2.5016
			(2.1364)	(2.1987)
13th century				0.4273
				(2.1678)
14th century				27.7798***
				(8.8441)
15th century				4.7241
				(3.5811)
16th century				-0.0156
				(2.5141)
17th century				0.9846
				(1.8061)
Century indicators	No	No	No	Yes
Clustered SE (family)	Yes	Yes	Yes	Yes
Observations	645	645	645	645
R^2	0.004	0.016	0.017	0.038
Adj. R^2	0.002	0.011	0.009	0.023

Notes: The dependent variable is the level difference in titles between older and younger brothers. Positive values indicate higher productivity for the older sibling. Standard errors clustered at the family level are reported in parentheses. Column (4) includes century indicator variables, with the eighteenth century used as the omitted reference category.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 15: Birth-Order Productivity Gap by Historical Period (Log Specification)

	(1) Pre-16th century	(2) 16th century onward
	Log difference in titles	
Constant	-0.450	-0.170
	(1.009)	(0.276)
Non-egalitarian inheritance	-0.453	0.211
	(0.914)	(0.248)
Unknown inheritance system	-1.568	0.155
	(2.551)	(0.283)
Age gap	0.174***	0.014
	(0.044)	(0.016)
Father in academia	0.829	-0.144
	(1.084)	(0.212)
Different institution	0.620	0.587**
	(1.021)	(0.247)
Century 13	-2.386**	—
	(0.923)	—
Century 14	1.049	—
	(0.919)	—
Century 16	—	0.082
	—	(0.307)
Century 17	—	0.159
	—	(0.222)
Century controls	Yes	Yes
Clustered SE (family)	Yes	Yes
Observations	33	612
R^2	0.341	0.015
Adj. R^2	0.156	0.004

Notes: The dependent variable is the logarithmic difference in titles between older and younger brothers. Positive values indicate higher productivity for the older sibling. All specifications include century controls. The omitted century category is the 15th century in Column (1) and the 18th century in Column (2). Standard errors clustered at the family level are reported in parentheses.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 16: Birth-Order Productivity Gap by Historical Period (Levels Specification)

	(1) Pre-16th century	(2) 16th century onward
	Difference in titles	
Constant	-1.246	-2.478
	(4.768)	(2.149)
Non-egalitarian inheritance	-12.663**	1.447
	(5.064)	(1.957)
Unknown inheritance system	-20.357	3.196
	(16.716)	(2.218)
Age gap	0.702**	0.168
	(0.333)	(0.108)
Father in academia	0.173	-0.905
	(7.440)	(1.701)
Different institution	20.117**	4.688**
	(8.241)	(2.044)
Century 13	-11.350	—
	(7.256)	—
Century 14	23.641**	—
	(10.286)	—
Century 16	—	0.041
	—	(2.518)
Century 17	—	0.954
	—	(1.803)
Century controls	Yes	Yes
Clustered SE (family)	Yes	Yes
Observations	33	612
R^2	0.494	0.018
Adj. R^2	0.352	0.007

Notes: The dependent variable is the raw difference in titles between older and younger brothers. Positive values indicate higher productivity for the older sibling. All specifications include century controls. The omitted century category is the 15th century in Column (1) and the 18th century in Column (2). Standard errors clustered at the family level are reported in parentheses.

Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

Table 17: Birth-Order Productivity Gap with Country Controls

	(1) *Asinh difference in titles*
Constant	-0.149
	(0.348)
Non-egalitarian inheritance	0.243
	(0.258)
Unknown inheritance system	-0.044
	(0.355)
Age gap	0.020*
	(0.011)
Father in academia	-0.143
	(0.187)
Different institution	0.388*
	(0.200)
13th century	0.064
	(0.280)
14th century	1.776***
	(0.577)
15th century	0.981***
	(0.349)
16th century	0.092
	(0.224)
17th century	0.140
	(0.167)
Country controls (reference: Germany (DEU))	
Austria (AUT)	-2.169***
	(0.423)
Belgium (BEL)	1.691*
	(1.021)
Belarus (BLR)	-1.783**
	(0.850)
Switzerland (CHE)	-0.172
	(0.461)
Denmark (DNK)	-0.572
	(0.487)
Spain (ESP)	0.483
	(0.380)

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	(1)
	Asinh difference in titles
Finland (FIN)	-0.484
	(0.738)
France (FRA)	-0.156
	(0.277)
United Kingdom (GBR)	-0.086
	(0.286)
Croatia (HRV)	2.341**
	(0.983)
Ireland (IRL)	0.447
	(0.823)
Italy (ITA)	0.018
	(0.349)
Lithuania (LTU)	2.226***
	(0.856)
Netherlands (NLD)	0.034
	(0.371)
Norway (NOR)	0.164
	(0.278)
Poland (POL)	-0.656
	(0.709)
Portugal (PRT)	-0.676
	(0.899)
Russia (RUS)	1.061**
	(0.511)
Slovakia (SVK)	1.151***
	(0.342)
Sweden (SWE)	-0.186
	(0.449)
Ukraine (UKR)	0.445
	(0.693)
United States (USA)	0.074
	(0.359)
Unknown country	-1.959***
	(0.313)
Century controls	Yes
Country categorical controls	Yes

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	(1)
	Asinh difference in titles
Clustered SE (family)	Yes
Observations	645
R^2	0.080
Adj. R^2	0.031

Notes: The dependent variable is the inverse hyperbolic sine (asinh) difference in titles between older and younger brothers. Positive values indicate higher productivity for the older sibling. The specification includes century indicator variables, with the eighteenth century used as the omitted reference category, together with country categorical controls constructed from the birthplace country identifiers of the brothers. Germany (DEU), the country with the largest number of observations, is used as the omitted reference category. Inheritance systems remain assigned at the regional NUTS level. Several countries contain only a very limited number of observations, reflecting the highly uneven geographical distribution of historically identifiable scholarly brother pairs. Standard errors clustered at the family level are reported in parentheses.
Significance levels: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.